

## FINAL EXPLANATION: MATHEMATICS GRADE 10

### *Student Assessment Document*

#### FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 2.2 AREA OF POLYGONS: FINAL EXPLANATION

##### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

Answer the driving question: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available? Remember the shamba left to three siblings in Nakuru County. It has five boundary corners marked by old fence posts, and no two adjacent sides are equal. Use what you learned in all eight lessons. Write your answers in the space under each prompt. Show all working for calculations, include units, and explain your reasoning in full sentences. Use your Growing Model and the Driving Question Board to help you.

#### Part 1: Why the usual formula is not enough

Surveyor Wanjiku stands at the edge of the five-sided shamba. She cannot simply use  $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$  on it. (a) Explain two reasons why this formula cannot be applied directly to the whole shamba. (b) What must be true about the 'height' in this formula? (c) What exactly did the class need to find out to solve the problem?

*Write your answer here:*

### Part 2: Choosing and using triangle formulas

Wanjiku's field notes contain two triangular sections. Triangle ABC:  $AB = 120$  m,  $AC = 95$  m, and the angle at A between them is  $68^\circ$ . Northern sub-plot: sides 40 m, 55 m and 63 m, with no angle or height recorded. (a) For each triangle, name the formula you would use and explain why it is the right choice. (b) Calculate the area of each triangle. Give answers to the nearest square metre. (c) Why must the angle in the sine formula be the included angle?

*Write your answer here:*

### Part 3: Breaking the five-sided plot into triangles

(a) How many triangles does the five-sided shamba break into when you draw diagonals from one vertex? Which diagonals would you draw from vertex A, and what are the triangles called? (b) State the rule for any polygon with  $n$  sides. (c) Explain how you would then find the total area, and say how you would choose the formula for each triangle.

*Write your answer here:*

### Part 4: Testing ideas and spotting mistakes

(a) A learner says: 'Longer sides always give a larger area, so a triangle with sides 60 m and 90 m must have a bigger area than any triangle with shorter sides, whatever the angle between the sides.' Rewrite this into a clear correct statement about two triangles that have the same two sides but different included angles. Then calculate the area of a triangle with sides 60 m and 90 m and an included angle of  $30^\circ$ , and explain what this shows about the effect of the included angle. (b) Why can the regular polygon formula  $A = ns^2 \div [4 \tan(180^\circ/n)]$  not be used on Wanjiku's shamba?

*Write your answer here:*

### Part 5: Your final explanation

**Answer the driving question in a complete explanation: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available? Your answer should (a) describe the steps the surveyor follows from start to finish, (b) name the formulas in her toolkit and say when each one is used, (c) explain how the area would be converted for practical land use, using Triangle ABC as an example, and (d) say how your thinking has changed since Lesson 1.**

Write your answer here:

| RUBRIC                                                                             |                                                                                                                                               |                                                                                                                     |                                                                                     |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Criterion                                                                          | Excellent (4)                                                                                                                                 | Proficient (3)                                                                                                      | Developing (1–2)                                                                    |
| Explaining the limitation of $\frac{1}{2} \times \text{base} \times \text{height}$ | Clearly explains that a perpendicular height is needed, that it cannot be measured on the shamba, and links this to the need for new methods. | States that the height is needed and cannot be measured but gives little link to the new methods.                   | Gives a vague or incorrect reason, such as confusing the height with a side length. |
| Selecting and applying formulas correctly                                          | Chooses the correct formula for each triangle with a justification based on the data, calculates correctly with units and sensible rounding.  | Chooses the correct formulas and calculates mostly correctly, with a small slip in working, units or justification. | Chooses the wrong formula or makes major calculation errors.                        |
| Decomposing polygons                                                               | States the $(n - 2)$ rule, names the diagonals and triangles for the shamba, and explains summing the areas.                                  | Gives the number of triangles and the summing idea but omits some detail, such as the diagonals or the rule.        | Does not decompose the polygon, or gives an incorrect number of triangles.          |
| Scientific explanation of the driving question                                     | Gives a complete, logical explanation connecting all steps, the toolkit, unit conversion and personal growth in understanding.                | Explains most steps and tools but leaves out unit conversion or reflection.                                         | Gives a list of formulas with no connection to the surveyor's problem.              |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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